TSM 098

Homework 5

Numerical fraction operations

Topics 19, 21-22 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Products, quotients, and common multiples

1. Find the LCM of each pair using prime factorization: (a) 12 and 18; (b) 8 and 20; (c) 9 and 15.

Multiply or divide

2. Compute and reduce completely. Show cancellation or the reciprocal step.

$\frac{-3}{8} \cdot \frac{4}{9}$

3. Compute and reduce completely. Show cancellation or the reciprocal step.

$\frac{14}{15} \cdot \frac{25}{21}$

4. Compute and reduce completely. Show cancellation or the reciprocal step.

$\frac{-5}{12} \cdot \frac{-18}{25}$

5. Compute and reduce completely. Show cancellation or the reciprocal step.

$\frac{7}{9} \div \frac{14}{15}$

6. Compute and reduce completely. Show cancellation or the reciprocal step.

$\frac{-4}{5} \div \frac{2}{3}$

7. Compute and reduce completely. Show cancellation or the reciprocal step.

$\frac{3}{5} \div \left(-\frac{9}{10}\right)$

Add or subtract

8. Compute using a common denominator and reduce.

$\frac{5}{6} + \frac{1}{4}$

9. Compute using a common denominator and reduce.

$\frac{7}{8} - \frac{1}{3}$

10. Compute using a common denominator and reduce.

$$-\frac{2}{5}+\frac{3}{10}$$

11. Compute using a common denominator and reduce.

$$\frac{4}{9}-\left(-\frac{1}{6}\right)$$

12. Compute using a common denominator and reduce.

$$\frac{7}{12}+\frac{5}{18}$$

13. Compute using a common denominator and reduce.

$$\frac{3}{8}-\frac{7}{20}$$

Mixed operations and explanations

14. Compute and reduce.

$$\frac{2}{3}-\frac{1}{4}\cdot\frac{8}{9}$$

15. Compute and reduce.

$$\left(\frac{5}{6}-\frac{1}{4}\right)\div\frac{7}{12}$$

16. Evaluate the complex fraction.

$$\frac{\frac{3}{4}+\frac{1}{6}}{\frac{5}{8}}$$

17. A student writes 1/2 + 1/3 = 2/5. Explain the denominator error and give the correct sum.

18. Explain why dividing 3/4 by 1/2 gives a result larger than 3/4. Compute the quotient and describe how many halves fit into three-fourths.